

HelixVPN — Unified Phase Roadmap

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Revision: 1 **Last modified:** 2026-07-04T12:00:00Z **Status:** active
— the single top-level index reconciling every phase document under docs/research/

Purpose. docs/research/ accumulated four generations of planning documents (mvp/, mvp2/, mvp3/, mvp_final/) written at different times, and — as documented honestly in §4 below — using **three different, mutually inconsistent phase-numbering schemes**. This document is the **single index** a new reader (or a new agent session) should start from: it lists every phase, what it delivers, where its documents live, how phases depend on each other, and it resolves the directory-naming-vs-phase-numbering confusion explicitly rather than silently. It does not re-specify any phase's content — every phase's authoritative content stays in its own document set, linked below.

Table of contents

- [1. The canonical phase list](#)
 - [2. Phase dependency graph](#)
 - [3. Per-phase entry/exit gate summary](#)
 - [4. Naming/structure reconciliation \(read this before anything else\)](#)
 - [5. Consolidated open decisions requiring operator input](#)
 - [Sources](#)
-

1. The canonical phase list

This is the numbering this project uses **going forward**, adopted specifically for this reconciliation (see §4 for why it differs from numbering already embedded in some existing documents):

Phase	One-paragraph description	Primary docs	Depends on
Phase 0 — Spike	Throwaway <i>bodies</i> on <i>production interfaces</i> , proving the hard technical risks before committing to the full build: the WireGuard data path through a gateway edge, one QUIC/MASQUE obfuscation mode, the iOS Network Extension memory ceiling (the reason the shared core is Rust, not Go), and a Go-vs-Rust edge-language benchmark. Nothing here ships to a user; the interfaces it proves (the Transport trait, the FFI surface, the reconciler) survive into Phase 1.	docs/research/mvp/final/06-phase0-spike-wbs.md; architecture context in docs/research/mvp/04_VPN_CLD/HelixVPN-Architecture-Refined.md §12	none (first phase)
	The self-hostable core:	docs/research/mvp/final/07-phase1-mvp-wbs.md;	Phase 0

Phase	One-paragraph description	Primary docs	Depends on
Phase 1 — Control-plane MVP	the Go modular-monolith control plane (helixd — identity, registry, ipam, pki, policy, coordinator, events, telemetry, api), Postgres with Row-Level Security and no durable connection/traffic table (schema-lint enforced), Redis Streams as the event backbone, the WatchNetworkMap push-based reconciliation protocol, the policy compiler (default-deny, AllowedIPs + verdict maps), helixvpctl, and rootless Podman quadlet deployment. One client platform (Linux or Android) proves the vertical slice end-to-end.	spine docs/research/mvp/final/SPECIFICATION.md §8.1	
Phase 2 — Client application suite	The 8-platform, 7-OS client application layer built on a	docs/research/mvp2/README.md (entry point), docs/research/mvp2/MVP2_OVERVIEW.md, docs/research/mvp2/	Phase 1

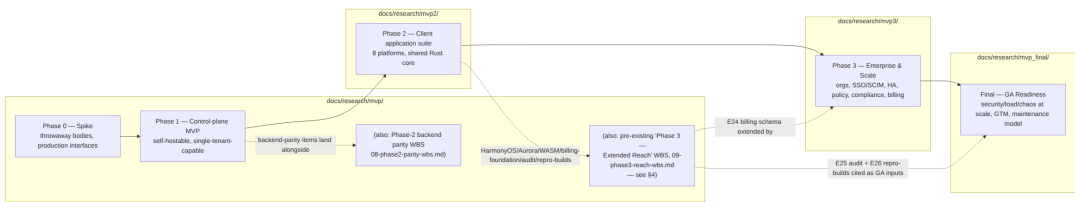
Phase	One-paragraph description	Primary docs	Depends on
Phase 3 — Enterprise & Scale	shared Rust core (helix-core, 70-85% code reuse): Tauri v2 desktop (macOS/Windows/Linux), Flutter mobile (Android/iOS/HarmonyOS), Qt6/QML (Aurora OS), and a browser extension + PWA companion for Web. Also covers, per the <i>pre-existing</i> Phase-2 backend-parity WBS, the full obfuscation transport set, DAITA traffic shaping, multi-hop, post-quantum handshake, and HA/multi-region groundwork on the control-plane side.	MVP2_ARCHITECTURE.md; backend-parity WBS docs/research/mvp/final/08-phase2-parity-wbs.md	
	Takes the working MVP + client suite to an enterprise-ready, horizontally-scalable, commercially deployable platform: organizations/	docs/research/mvp3/MVP3_ENTERPRISE_SCALE.md	Phase 2 (client suite Tier-1 platforms working)

Phase	One-paragraph description	Primary docs	Depends on
	teams/ permission-based RBAC above the existing tenant model, enterprise SSO (SAML alongside OIDC) + SCIM provisioning + device- posture/ conditional- access, multi- region HA hardening (resolving open questions left in the Phase-2 HA design), an advanced policy engine (time/context- aware conditions, per-app split- tunnel templates, destination- category egress controls, SIEM export), compliance- grade observability (tamper- evident audit retention, a SOC2/ ISO27001/ GDPR control map), marketplace/ billing, and advanced		

Phase	One-paragraph description	Primary docs	Depends on
	networking (site-to-site mesh, dual-stack IPv6, QoS, hardened DPI-resistance). The last phase before General Availability: a follow-up security audit scoped to the Phase-3 enterprise surface, load/chaos testing at commercial scale, four-audience documentation completeness, an on-call/incident-response process, go-to-market readiness (pricing, legal, 8 app-store listings, onboarding), and the long-term maintenance model (LTS/versioning policy, deprecation policy, FFI/API backward-compatibility guarantees, security-patch SLA).		
Final — GA Readiness	process, go-to-market readiness (pricing, legal, 8 app-store listings, onboarding), and the long-term maintenance model (LTS/versioning policy, deprecation policy, FFI/API backward-compatibility guarantees, security-patch SLA).	docs/research/mvp_final/MVP_FINAL_GA_READINESS.md	Phase 3

docs/research/mvp4/TBD.md exists as an empty placeholder and is **out of scope for this reconciliation** — it has no defined content yet and is not one of the phases this document indexes; a future pass should either populate it with a genuinely new phase or remove it as redundant once Phase 3/Final absorb its intended scope (this is itself an open item, §5).

2. Phase dependency graph



3. Per-phase entry/exit gate summary

Phase	Entry condition	Exit condition (all must hold, with captured evidence)
Phase 0	none	Gates G1-G6 pass: $\geq 80\%$ bare-link WG throughput; MASQUE survives a DPI UDP block at $\geq 50\%$ of plain-WG throughput; iOS NE memory ceiling proven with $\geq 30\%$ headroom; Go-vs-Rust edge benchmark decided; FFI drives the core end-to-end from Dart; push-based reconcile converges with zero polling.
Phase 1	Phase 0 gates green	The 8-criterion MVP Definition of Done (mvp/final/07-phase1-mvp-wbs.md): self-host from zero, enroll connector+client, authorized reach proven + unauthorized denied, auto-escalation to MASQUE under a block, policy edit reflected $< 1s$, device revoke $< 1s$, kill-switch+DNS-leak proven, no durable connection log exists.
Phase 2	Phase 1 DoD green	Full Mullvad feature-parity matrix green; desktop/mobile/Aurora/HarmonyOS/Web builds pass the same 8 MVP DoD criteria on-device; multi-hop + P2P/NAT-traversal + PQ handshake interop

Phase	Entry condition	Exit condition (all must hold, with captured evidence)
Phase 3	Phase 2 Tier-1 client suite working	proven; HA failover <30s with transport preserved (per the <i>pre-existing</i> Phase-2 backend WBS, 08-phase2-parity-wbs.md), and Tier-1 client platforms (macOS/Windows/Linux/Android) have a working build per mvp2/MVP2_OVERVIEW.md §4.2. All gates G30–G37 in mvp3/MVP3_ENTERPRISE_SCALE.md §10.2 green: org/RBAC model live with RLS intact; SAML+SCIM+posture E2E proven; multi-region HA chaos-tested; advanced policy engine (conditions+templates+SIEM export) proven; tamper-evident audit + control mapping delivered; billing/marketplace rollup + reseller model proven; site-to-site/dual-stack/QoS proven; full-suite retest with zero no-logging-as-code (S6/S7) regression.
Final (GA)	Phase 3 gates green, and the pre-existing mvp/final/09-phase3-reach-wbs.md gates G20–G26 at least pass or honestly pending_device/operator_blocked	All gates G40–G44 in mvp_final/MVP_FINAL_GA_READINESS.md §5.2 green: follow-up security audit zero unresolved Critical/High; commercial-scale load+chaos evidence captured; four-audience docs complete; on-call+game-day proven; pricing/legal/8-store-listings/onboarding complete; LTS/deprecation/compat-matrix/patch-SLA published; operator explicit go/no-go recorded.

4. Naming/structure reconciliation (read this before anything else)

The directory layout under docs/research/ does **not** number phases 1:1 with directory suffixes, and the corpus that accumulated over time actually contains **three different, mutually inconsistent phase-numbering narratives**. This section names all three, states which one is canonical going forward, and cross-references the others without silently rewriting them (per this project's own no-guessing/no-silent-

renumbering discipline — a directory rename or content rewrite is a visible, deliberate operator decision, not something a documentation pass does quietly).

4.1 Narrative A — docs/research/mvp/final/SPECIFICATION.md's internal Phase 0-3

The most detailed and most-referenced narrative, embedded in the `mvp/` directory's own spine document (§8) and its per-phase WBS docs:

- Phase 0 — Spike
- Phase 1 — MVP (self-hostable)
- **Phase 2 — Parity + Reach:** full transport set, DAITA, multi-hop, PQ handshake, **desktop apps** (Windows/macOS), policy-as-code/GitOps, **HA + multi-region**. Note this is *backend/control-plane feature completion*, not the full 8-platform client suite.
- **Phase 3 — Extended Reach:** HarmonyOS NEXT + Aurora OS native tunnel shims, an in-browser WASM MASQUE proxy, **billing-optional multi-tenant** Console + audit, third-party security audit + reproducible builds. Detailed in `docs/research/mvp/final/09-phase3-reach-wbs.md`.

4.2 Narrative B — docs/research/mvp2/MVP2_OVERVIEW.md §3.1's internal Phase 1-4

A second, independently-written narrative embedded in the MVP2 overview's own "Project Context" section:

- Phase 1 (MVP1) — "Server Infrastructure & Admin Backend" — marked **[COMPLETED]**
- Phase 2 (MVP2) — "Client Applications (8 platforms)" — **"THIS DOCUMENT"**
- Phase 3 (MVP3) — "Enterprise Features & Ecosystem" — **[PLANNED]**
- Phase 4 (MVP4) — "Advanced Anti-Censorship & Post-Quantum" — **[PLANNED]**

This narrative is flagged here as internally inconsistent with the actual project state, not resolved. Its Phase 1 description ("VPN Server Network: globally distributed WireGuard server fleet," "Billing Integration: subscription management... **[COMPLETED]**") does not match the actual Phase 0/1 specification in `mvp/final/SPECIFICATION.md` (a single self-hostable control-plane MVP with billing explicitly out of scope until the Phase-3 "billing-optional" epic, and no claim of a globally-distributed managed server fleet). This reads as boilerplate carried over from a generic SaaS-VPN framing rather than a description of *this* project's actual, spec-only, not-yet-built Phase 0/1. It is called out here for

operator awareness. It is **not corrected in this pass** — docs/research/mvp2/ is a parallel workstream's document set and is explicitly out of scope for this reconciliation's edits (per the constraint that this reconciliation only creates new files and does not modify mvp/ or mvp2/).

4.3 Narrative C — this document's canonical numbering (§1 above)

The numbering this reconciliation adopts, matching the directory-to-phase mapping the operator specified for this work: mvp/ = Phase 0 + Phase 1, mvp2/ = Phase 2, mvp3/ = Phase 3, mvp_final/ = Final/GA. **This is the numbering used going forward** for any new cross-phase document (this one, and the two it indexes).

4.4 The concrete collision, and how it is resolved without duplication

Narrative A's "Phase 3 — Extended Reach" and this reconciliation's "Phase 3 — Enterprise & Scale" (mvp3/MVP3_ENTERPRISE_SCALE.md) are **both labeled Phase 3**, and they are **genuinely different in scope**:

	Narrative A's "Phase 3" (mvp/final/09-phase3-reach-wbs.md)	This reconciliation's "Phase 3" (mvp3/MVP3_ENTERPRISE_SCALE.md)
Scope	Platform reach (HarmonyOS/ Aurora native shims, WASM browser proxy) + hardening (third-party audit, reproducible builds) + a billing-optional multi-tenant <i>foundation</i>	Organizational scale: orgs/ teams/RBAC, enterprise SSO/ SCIM/device-trust, multi-region HA hardening, advanced policy engine, compliance observability, full marketplace/billing, advanced networking
ID namespace	HVPN-P3-NNN, gates G20-G26	HVPN-ENT-NNN, gates G30-G39 (deliberately distinct, no collision)
Relationship	—	Extends, never duplicates: mvp3/'s billing model (§3.4, §8) is built directly on top of Narrative A's HVPN-P3-E24 billing schema (plan/ subscription/usage_counter); mvp3/'s decision register cites

Narrative A's "Phase 3" (mvp/final/09-phase3-reach-wbs.md)	This reconciliation's "Phase 3" (mvp3/MVP3_ENTERPRISE_SCALE.md)
Narrative A's HVPN-P3-E25 audit epic rather than re-scoping it	

This reconciliation's recommendation (an open decision, not silently applied — see §5): treat Narrative A's Phase 3 as a **platform-completion and hardening track** that runs alongside — and is a prerequisite input to — this reconciliation's Phase 3 (enterprise/scale) and Final (GA) phases, rather than renaming either document set. The dependency graph in §2 shows this explicitly: Narrative A's WBS feeds artifacts (the billing schema, the audit report, the reproducible-build attestation) into both mvp3/ and mvp_final/, without either document needing to be rewritten. If the operator later prefers a clean single numbering (e.g. relabeling Narrative A's Phase 3 as "Phase 2c" or "Phase 2.5," since HarmonyOS/Aurora/WASM-proxy are genuinely deferred *client-platform* work), that is a deliberate rename tracked as an open item (§5) — not something this document performs unilaterally.

5. Consolidated open decisions requiring operator input

Pulled together from both new phase documents, plus the reconciliation work above, so an operator reviewing this roadmap sees every outstanding decision in one place:

#	Decision	Source	Recommendation
R-1	Whether to rename/renumber Narrative A's "Phase 3 — Extended Reach" (mvp/final/09-phase3-reach-wbs.md) to remove the Phase-3 label collision described in §4.4	this document, §4.4 (also mvp3/MVP3_ENTERPRISE_SCALE.md D-ENT-1)	keep both cross-referenced as-is for now; revisit if the label collision causes real confusion once a unified docs/workable_items.db is populated with both HVPN-P3-* and HVPN-ENT-* rows
R-2	Whether/how to correct mvp2/	this document, §4.2	flagged, not corrected here

#	Decision	Source	Recommendation
	MVP2_OVERVIEW.md §3.1's inconsistent "MVP1 completed / global server fleet / billing already live" framing (§4.2)		(out of scope for this reconciliation pass — belongs to the mvp2/ workstream owner)
R-3	Pricing model for the subscription tiers	mvp3/ MVP3_ENTERPRISE_SCALE.md §11 D-ENT-2; mvp_final/ MVP_FINAL_GA_READINESS.md §7 D-GA-1	not specified — highest-priority business decision blocking GA go-to- market work
R-4	Which compliance certification (SOC2, ISO27001, or neither) to pursue, and whether before or after GA	mvp3/ §11 D-ENT-3; mvp_final/ §7 D-GA-4	not specified — depends on target market (US enterprise vs. EU enterprise vs. self- host-only)
R-5	Whether the organization layer above tenant (mvp3/ §3.1) is needed for the first enterprise release, or can trail RBAC/ SSO/HA	mvp3/ MVP3_ENTERPRISE_SCALE.md §11 D-ENT-4	recommend shipping RBAC+SSO+HA first; org/reseller layer can trail
R-6	DLP posture — confirm "metadata/ category-only egress control, no payload inspection" as final product positioning	mvp3/ MVP3_ENTERPRISE_SCALE.md §6.3, §11 D-ENT-5	recommend accepting the boundary; payload inspection contradicts the no- logging-as-code architecture
R-7	SLA percentages and security- patch-SLA time windows	mvp3/ §11 D-ENT-6; mvp_final/ §7 D-GA-3	draft numbers proposed in both documents; need operator/legal confirmation before publishing

#	Decision	Source	Recommendation
R-8	LTS cadence	mvp_final/ MVP_FINAL_GA_READINESS.md §7 D-GA-2	recommend calendar-based cadence; not decided
R-9	GA launch sequencing across the 8 client platforms (simultaneous vs. staged by platform- readiness)	mvp_final/ §7 D-GA-5	recommend staged, gated by the HarmonyOS/ Aurora device- proof gates (G21/ G22) already in mvp/ final/09-phase3- reach-wbs.md
R-10	Disposition of docs/research/ mvp4/TBD.md (empty placeholder, no defined scope, not indexed by this roadmap)	this document, §1	either populate with a genuinely new phase or remove once Phase 3/Final absorb its intended scope

Sources

- docs/research/mvp/final/SPECIFICATION.md — Narrative A's spine (§4.1).
- docs/research/mvp/final/06-phase0-spike-wbs.md, 07-phase1-mvp-wbs.md, 08-phase2-parity-wbs.md, 09-phase3-reach-wbs.md — the four existing WBS documents cross-referenced throughout.
- docs/research/mvp2/README.md, docs/research/mvp2/MVP2_OVERVIEW.md — Narrative B (§4.2), the Phase-2 client-suite platform matrix (§1, §3).
- docs/research/mvp3/MVP3_ENTERPRISE_SCALE.md — this reconciliation's Phase 3 (§1, §4.4).
- docs/research/mvp_final/MVP_FINAL_GA_READINESS.md — this reconciliation's Final/GA phase (§1).
- docs/research/mvp3/TBD.md, docs/research/mvp4/TBD.md, docs/research/mvp_final/TBD.md — the placeholder files found at the start of this work; mvp3/'s and mvp_final/'s scope is now defined by the documents above, mvp4/ remains an open item (§5, R-10).

End of unified phase roadmap.